PYTHON GUIDE

A sample searchable PDF for your JARVIS AI assistant

Purpose
Place this PDF inside your **JARVIS_AI/documents** folder, then ask JARVIS questions about topics in this guide. It gives you new content to test PDF text extraction and semantic retrieval.

Topics in this guide

- Functions, parameters, and return values
- Handling errors with try and except
- Reading text files safely
- Python classes and objects

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1. Functions, Parameters, and Return Values

A function is a named, reusable block of code. Parameters are names in a function definition that receive values. Arguments are the actual values supplied when calling the function. The **return** statement sends a result back to the caller.

Example

```
def add_numbers(first, second):
    return first + second

result = add_numbers(3, 4)
print(result) # 7
```

In this example, **first** and **second** are parameters. The values 3 and 4 are arguments. The function returns 7, which is stored in the variable result.

2. Handling Errors with try and except

An exception is an error detected while a program is running. A try block contains code that might fail. An except block handles a specified exception so the program can respond instead of stopping unexpectedly.

Example

```
try:
    age = int(input("Enter your age: "))
except ValueError:
    print("Please enter a whole number.")
```

The conversion int(...) raises a ValueError when the input cannot be converted into an integer. The except block handles that case.

3. Reading a Text File Safely

Python can read files using open(). Using a with statement is recommended because it closes the file automatically after the indented block finishes, even if an error occurs.

Example

```
with open("notes.txt", "r", encoding="utf-8") as file:
    content = file.read()

print(content)
```

The mode "r" means read. UTF-8 is an encoding commonly used for text. The file named notes.txt must exist in the location given to open().

4. Python Classes and Objects

A class defines the structure and behavior of a type of object. An object is an instance created from a class. The __init__ method is commonly used to initialize an object when it is created. The parameter self refers to the current instance.

Example

```
class Student:
    def __init__(self, name):
        self.name = name

student = Student("Abdul")
print(student.name) # Abdul
```

In this example, Student is the class and student is an object. The name attribute stores the value passed when the object is created.

Suggested JARVIS test questions

- What is the difference between a parameter and an argument?
- How does a try and except block work?
- Why is the with statement useful when reading a file?
- What are a class and an object in Python?